BECE MOCK EXAM
SCIENCE 2 & 1
Essay and Objective
2 hours 10 minutes

2 & 1

Name: .....

Index Number: .....

# Basic Education Certificate Mock Examination

SCIENCE 2 & 1
Essay and Objective

2 hours 10 minutes

*Do* **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in* **ink** *in the spaces provided above.*

*This booklet consists of* **two** *papers. Answer Paper* **2** *which comes first, in your answer booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last 1 hour 25 minutes after which the answer booklet will be collected. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last 45 minutes.*

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PAPER 2
ESSAY
[100 marks]

*This paper is in two sections:* **A** *and* **B**. *Answer Question* **1** *in Sections* **A** *and any other three questions in Sections* **B**.

*Answer all the questions in your answer booklet.*

*Credit will be given for clarity of expression and orderly presentation of material.*

SECTION A
[40 marks]

*Answer* **all** *the questions in this section.*

**1.** (a) **Fig. 1(a)** below is an illustration of carbon cycle showing unnamed processes labelled **A, B, C, D, E, F** and **G.**
*Study the figure carefully and answer the questions that follow.*

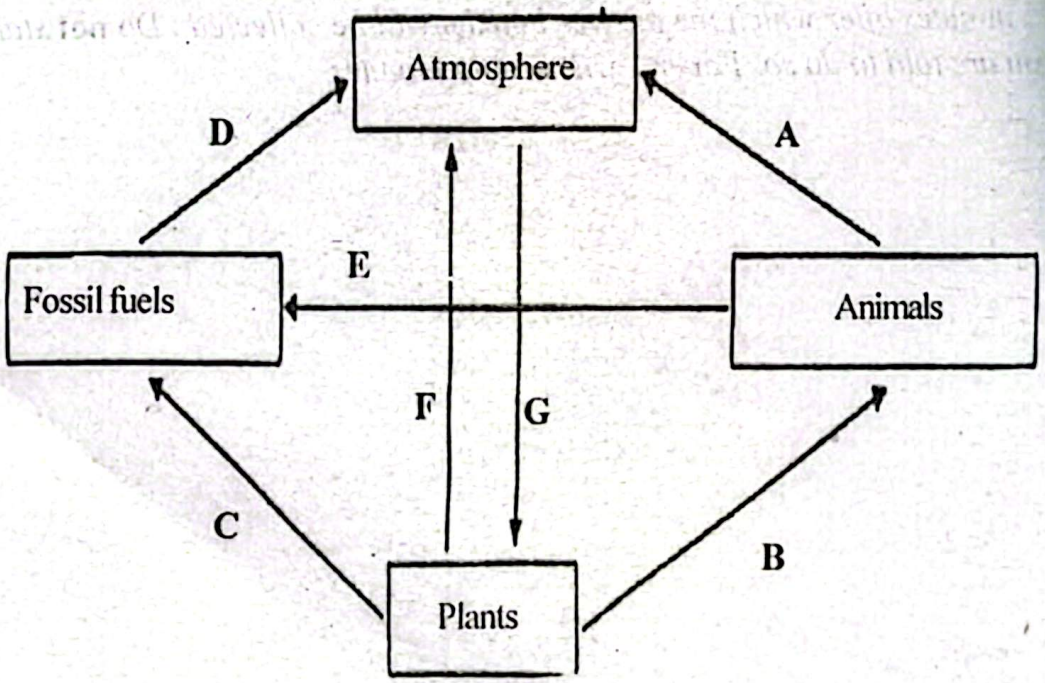



Fig. 1(a)

(i) Identify the processes labelled **A, B, C, D, E, F** and **G.**
(ii) State **two** environmental effects of disrupting the carbon cycle.
(iii) Briefly explain the carbon cycle. [10 marks]

(b) **Fig. 1(b)** below is an illustration of mode of heat transfer.
*Study it carefully and answer the questions that follow.*

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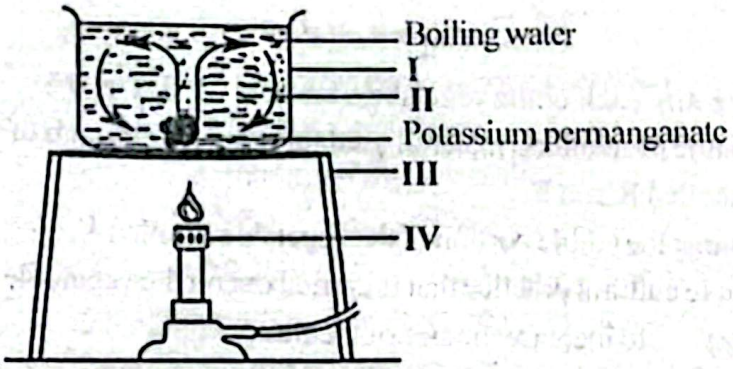



Fig. 1(b)

(i) State the aim of the experiment.

(ii) Name the parts labelled I, II, III and IV.

(iii) State what would happen if the set-up is left for one hour?

(iv) Give two importance of the heat transfer demonstrated above.

[10 marks]

(c) **Fig.1(c)** below is an illustration to demonstrate how matter changes from one state to another. *Study the diagram and answer the questions that follow.*

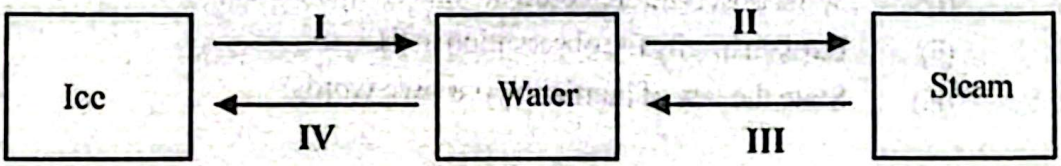



Fig. 1 (c)

(i) State what takes place in each of the stages labelled I, II, III and IV.

(ii) In which of the labelled stages is heat added?

(iii) In which of the labelled stages is heat removed?

(iv) What happens to the temperature in:

($\alpha$) stage I;

($\beta$) stage IV.

(v) State how the particles in ice are arranged? [10 marks]

(d) **Fig. 1(d)** below is an illustration of various vegetable crops.
*Study the illustration carefully and answer the questions that follow.*

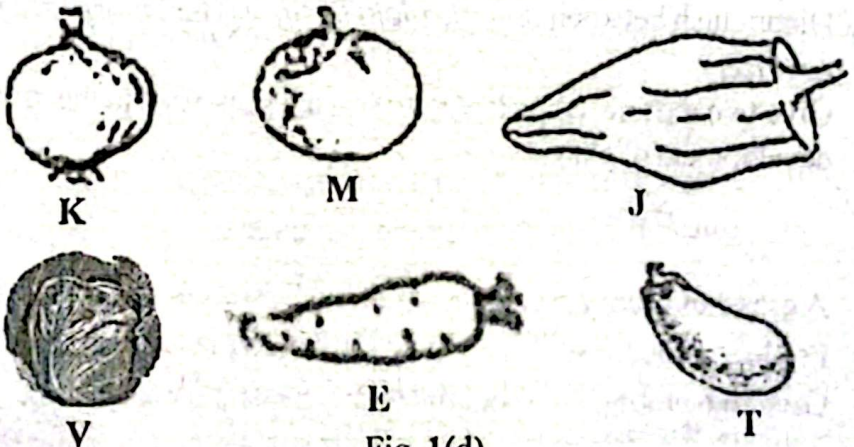



Fig. 1(d)



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(i) Identify **each** of the vegetables labelled **M, J, T** and **V**.

(ii) Name the planting material used for propagating **each** of the vegetables labelled **K** and **E**.

(iii) Name the edible portion of the vegetable labelled **V**.

(iv) State cultural practice that is carried out on the vegetable labelled **M**;

($\alpha$) to increase the size of the fruit;

($\beta$) for clean fruit to be harvested. [10 marks]

## SECTION B
[60 marks]

*Answer* **three** *questions* **only** *from this section.*

(a) Amuzu placed a paper on a cup and a coin on the paper. He quickly pulled the paper out.

(i) What observation would be made about the coin?

(ii) Explain briefly the observation in (i).

(iii) State the law of inertia in your own words. [5 marks]

(b) (i) Explain *malnutrition.*

(ii) State the cause(s) of the following diseases:

($\alpha$) ricket;

($\beta$) pellagra;

($\theta$) goitre. [5 marks]

(c) (i) Distinguish between the following:

($\alpha$) macro-nutrients and micro-nutrients;

($\beta$) ruminants and non-ruminants.

(ii) Name **two** breeds of rabbit. [6 marks]

(d) (i) Distinguish between *organic plant fertilizers* and *inorganic plant fertilizers.*

(ii) Give **two** reasons why plant nutrients are important to the growth and development of plants. [4 marks]

(a) (i) A grassland ecosystem consists of toads, snakes and grasshoppers. Produce a food chain using the information provided above.

(ii) Give **two** abiotic components of the grassland ecosystem. [4 marks]

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(b) (i) Explain the term *green economy*.

(ii) Differentiate between the following waste disposal practices:

($\alpha$) landfill and incineration;

($\beta$) recycling and plasma gasification. [4 marks]

(c) (i) Compare natural gas to biogas as sources of energy.

(ii) Give **three** differences between the two sources of energy.

(iii) Describe how to produce biogas. [7 marks]

(d) (i) Name the life process involved in **each** of the following daily activities:

(á) withdrawal of the hand from a hot cooking pot in the kitchen;

(â) increase in the height of a maize seedling in the garden.

(ii) A football player runs after the football on the school field. Name **three** systems of the human body involved in this activity. [5 marks]

4. (a) (i) List **two** plant parts, other than seeds, that can be used to plant crops.

(ii) State **two** advantages of planting different crops on different seedbeds.

(iii) State **one** problem that may occur if yam is planted on flat seedbed. [5 marks]

(b) (i) Explain briefly the term *cell organelle*.

(ii) Give **one** function of **each** of the following organelles:

($\alpha$) golgi body;

($\beta$) chloroplast;

($\theta$) cell membrane. [5 marks]

(c) A facilitator visited Akosombo Hydroelectric Dam with his class. The tour guide explained how energy changes from one form to another at the dam.

(i) Draw a simple energy flow diagram to show how energy changes from one form to another at the dam.

(ii) Name **two** factors that affect the water's energy stored behind the dam. [6 marks]

(d) (i) Give **two** characteristics of homogeneous mixture.

(ii) Distinguish between *atomic number* and *mass number*. [4 marks]

5. (a) (i) State **two** characteristics of viral diseases.

(ii) List **three** symptoms of swine flu disease. [5 marks]

(b) (i) Define the term *work input*.

(ii) An effort of 60 N is applied onto a 2-metre long rigid bar to lift a box weighing180 N. The effort distance is 1.6 m. Calculate the:



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(α) load distance;

(β) efficiency of the machine. [5 marks]

(c) (i) Give **two** differences between *solute* and *solvent*.

(ii) Name the appropriate solvent for **each** of the following solutes:

(α) glucose;

(β) rubber. [4 mark]

(d) (i) Explain briefly how plants absorb water from the soil.

(ii) State **one** benefit of **each** of the following properties to crop production:

(α) soil texture;

(β) soil water content. [6 marks]

***END OF ESSAY TEST***

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*Answer all* **the** *questions.*

*Each question is followed by four options lettered* **A** *to* **D**. *Find the* **correct** *option for each question and shade* **in pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

Which of the following substances in **not** an element?

A. Aluminium
B. Ammonia
C. Oxygen
D. Sodium

*The correct answer is Ammonia, which is lettered B and therefore answer space B would be shaded.*

[ A ] [ B ] [ C ] [ D ]

*Think carefully before you shade the answer space; erase completely any answer you wish to change.*

*Do* **all** *rough work on this question paper.*

*Now answer the following questions.*

1. Which of the following tools is used for site selection and surveying of farmland?
   A. Mattock
   B. Cross-staff
   C. Headpan
   D. Sickle

2. A disease that spreads across many countries is called a
   A. epidemic.
   B. pandemic.
   C. endemic.
   D. outbreak.

3. The body organ that is **mainly** affected by COVID-19 is
   A. liver.
   B. kidney.
   C. heart.
   D. lungs.

4. The emission of greenhouse gases such as carbon dioxide into the atmosphere is a disadvantage of
   A. thermal energy.
   B. solar energy.
   C. tidal energy.
   D. hydroelectric energy.

5. Which of the following metals does **not** react with water?
   A. Aluminium
   B. Potassium
   C. Lithium
   D. Zinc

6. The **best** way of maintaining tools with moving parts is
   A. greasing the movable parts.
   B. washing them after use.
   C. using them for the right task.
   D. keeping them away from termites.

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7. Which of the following factors should be considered when selecting a site for vegetable crop production?

I. Method of propagation
II. Nature of land
III. Climatic factor

A. I and II only
B. II and III only
C. I and III only
D. I, II and III

8. Growing maize and rearing goat on the same piece of land is called
A. organic farming.
B. mixed farming.
C. inter-farming.
D. multiple farming.

*Use the organelles below to answer Questions* **9** *to* **11.**

I. Mitochondrion
II. Chloroplast
III. Nucleus
IV. Vacuole

9. The organelle which stores genetic information is
A. I.
B. II.
C. III.
D. IV.

10. The largest organelle in a plant cell is
A. I.
B. II.
C. III.
D. IV.

11. The organelles which are in animal cells are
A. I and III only
B. I, II and IV only
C. I, III and IV only
D. II, III and IV only

12. Which of the stages of mosquito is **commonly** referred to as tumblers?
A. Adult
B. Egg
C. Larvae
D. Pupae

13. Which of the following is an example of feed supplement?
A. Groundnut cake
B. Rice
C. Sorghum
D. Grass

14. Which of the following is a breed of pig?
A. Ashanti dwarf
B. West African dwarf
C. N'dama
D. Sahelian

15. A crane moves a load of 30 N through a distance of 5 m. If it uses an effort of 20 N to move through a distance of 3 m, calculate the work input.
A. 60 J
B. 90 J
C. 100 J
D. 150 J

16. The quantity of feed allowed on animals for specific period of time is
A. basal feed.
B. fodder.
C. ration.
D. hay.

17. The **major** nutrients supplied by leafy vegetables include all the following **except**
A. fats and oils.
B. proteins and minerals.
C. vitamins and mineral salts.
D. vitamins and proteins.



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18. When a force is applied on a body, it may result in all the following changes **except** a change in
A. speed of the motion of the body.
B. shape of the body.
C. direction of motion of the body.
D. volume of the body.

19. Which of the following is a physical property of clay soils?
A. They are easily drained
B. Pore spaces are large at medium
C. Surface cracks during the dry season
D. Soil particles have rough texture

20. The angle between a ray of light and the surface of a plane mirror it strikes is 35°. The angle of its reflection will be
A. 35°.
B. 45°.
C. 55°.
D. 60°.

21. Which of the following tools is an example of third class lever?
A. Claw hammer
B. Pair of tongs
C. Wheelbarrow
D. Pair of scissors

22. Which of the following is **not** a factor of climate change variation?
A. Altitude
B. Topography
C. Latitude
D. Weathering

23. A catapult may be used to kill a bird because its rubber band has
A. kinetic energy which is converted to potential energy.
B. kinetic energy which is converted to heat energy.
C. potential energy which is converted to kinetic energy.
D. potential energy which is converted to light energy.

24. Damaged tissues of the human body are repaired by
A. carbohydrates.
B. lipids.
C. vitamins.
D. proteins.

25. Which of the following farming systems can lead to total crop failure?
A. Crop rotation
B. Mixed farming
C. Mixed cropping
D. Mono cropping

26. When Benedict's solution is used to test for sugar, the colour changes to
A. red.
B. violet.
C. blue black.
D. brick-red.

27. Nitrogen molecule is represented by the chemical symbol $N_2$. This symbol represents two
A. ions of nitrogen.
B. atoms of nitrogen.
C. elements of nitrogen.
D. molecules of nitrogen.

28. In the digestive system of mammals, fatty acid and glycerol are absorbed in the
A. colon.
B. duodenum.
C. ileum.
D. stomach.

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29. A characteristic feature of a goat that distinguishes it from a sheep is the possession of
A. long tail.
B. four-chambered stomach
C. curled horns.
D. short erect tail.

30. Which of the following factors should **not** be considered when preparing feed for a farm animal?
A. Age of animal
B. Colour of the animal
C. Nutrient requirement of the animal
D. Palatability of feed

31. The **main** function of large intestine in the digestive system of humans is to
A. store faeces.
B. absorb water from undigested food.
C. absorb food nutrients from the digested food.
D. digest cellulose in the diet.

32. The relative proportion of various particle sizes in a given soil sample is
A. coarseness.
B. profile.
C. structure.
D. texture.

33. According to Newton's second law of motion, if mass is constant, increasing the force will
A. decrease acceleration.
B. increase acceleration.
C. have no effect.
D. stop motion.

34. In a charging capacitor, current is highest at the
A. end of charging.
B beginning of charging.
C. middle of charging.
D. during discharging.

35. A nutrient source with large particle size is **best** applied by
A. foliar spray.
B. side dressing.
C. ring method.
D. broadcasting.

36. Which of the following is an igneous rock?
A. Granite
B. Limestone
C. Sandstone
D. Shale

37. Which of the following is an example of a complex machine?
A. Wheelbarrow
B. Cutlass
C. Bicycle
D. Tweezers

38. Which of the following mixtures is a colloid?
A. Common salt in water
B. Sugar in water
C. Cooked starch in water
D. Alum in water

39. Distance travelled in a specified direction is known as
A. acceleration.
B. displacement.
C. speed.
D. velocity.

40. Low haemoglobin can be associated with insufficient intake of
A. protein.
B. glucose.
C. calcium.
D. iron.

*END OF PAPER*

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